

Room Acoustics Affects Immersion and Musical Behavior in Virtual Renderings of Real-World Spaces

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ABSTRACT

Virtual reality (VR) can not only be used to create fantastical environments and experiences, but also to simulate reality. An under-explored area of research is how room acoustics affects users' behavior and immersion in virtual-renderings of real spaces. Previous research has shown that musicians adapt their performance according to the reverberation of the space they are in, and that auditory feedback can affect immersion in virtual environments. This study investigates how room acoustics affects individuals' sense of immersion and musical behavior in VR. Participants performed short musical compositions in virtual-renderings of real-world spaces. We systematically varied the auditory feedback to simulate dry (no room acoustics) or reverberant conditions both with the same visuals. Our results revealed that the presence of room acoustics enhanced participants' sense of immersion and affected their playing-speed and key-press intensity. These findings highlight the importance of incorporating room acoustics into VR-design to foster immersion and influence user interaction.

Index Terms: music performance, virtual reality, virtual acoustics, multisensory integration, binaural, audition, vision.

1 INTRODUCTION

Extended reality technologies provide a great opportunity for researchers to create more ecologically valid experimental scenarios, without compromising on rigorous control. Virtual reality (VR) specifically, offers both experimental control and realism for studying real-world behavior in different manipulated conditions. For example, VR can be used to investigate how different acoustic conditions, such as ambient noise, affect cognitive performance [1]. It can also be used to learn more about music performance. Previous research has employed VR as an intervention to address music performance anxiety through exposure training [2] and as a tool for alleviating anxiety pre-performance [3]. However, for VR to be a useful research tool, it should mimic the real-world situation as accurately as possible. This does not necessarily mean an exact audio-visual replica of the real world. For a VR research tool to be effective, it needs to simulate the scenario in such a way that users/participants think and behave as they would in the real world. One way to encourage real-world behavior in VR is to focus on increasing participants' immersion during the virtual experience.

Immersion is a widely sought-after quality of any virtual experience. Developers want users to feel immersed in a space, as such experiences foster embodiment, presence, engagement with the environment, and ultimately, enjoyment [4, 5]. The design of virtual environments is often vision-driven. However, many researchers

argue that auditory feedback is just as important for fostering immersion in virtual environments [6, 7, 8]. When individuals participated in an experiment where they were placed in virtual environments with only sound or visuals followed by fully audio-visual environments, they not only preferred the multi-modal experiences but rated them as more immersive [9]. We can use these immersive audiovisual experiences to test real-world perception and behavior. Participants in a sound localization task performed similarly in real spaces versus the virtual recreations of the same spaces, indicating that realistic auditory feedback in virtual environments can drive user behavior [10].

Sound carries spatial cues that visuals alone cannot convey. When accurately modeled and rendered, room acoustics can contribute to an individual's sense of presence in a virtual space [11]. Technologies such as ray tracing, convolution reverb, and machine learning make it possible to simulate sound propagation in any given space, enabling believable auditory environments [12, 13]. These tools are specifically useful for recreating virtual renderings of real spaces that can then be used in scientific research, specifically related to room acoustics.

One area of human interactive behavior that is notably affected by room acoustics is music performance. Previous research has shown that musicians adapt their playing to different acoustic environments [14, 15, 16, 17]. Although these adaptations are mainly individualized and vary per person, there are some overall trends with regards to tempo. Participants tend to decrease their playing tempo in more reverberant conditions, such as those representing bigger physical spaces [16]. Based on the temporal differences resulting from room acoustics, we hypothesize that the intensity of participants' keyboard key presses will also be affected by reverberation.

This Study

This study aims to investigate how the type of auditory feedback (dry - without room acoustics versus reverberant - with room acoustics) influences user behavior and sense of immersion in virtual renderings of real-world spaces. We conducted an experiment in which participants performed two different musical compositions (scores) on an electronic keyboard in six different virtual reality environments. The visuals were presented using a head-mounted display (HMD), while auditory feedback was delivered binaurally via headphones. The experiment featured three virtual recreations of real-world music performance spaces: a rehearsal hall, a concert hall, and a church. The participants performed in each space under two conditions: dry and reverberant. In the dry conditions, participants saw the virtual performance space but heard a sound without any reverberation. In the reverberant condition, the auditory feedback aligned with the acoustic characteristics of the displayed space. To minimize order effects, the sequence of virtual environments was randomized, although all participants completed the dry conditions first to ensure consistent progression throughout the experiment. Participants also completed a short questionnaire at the end of the experiment to measure their sense of immersion in the different scenes.

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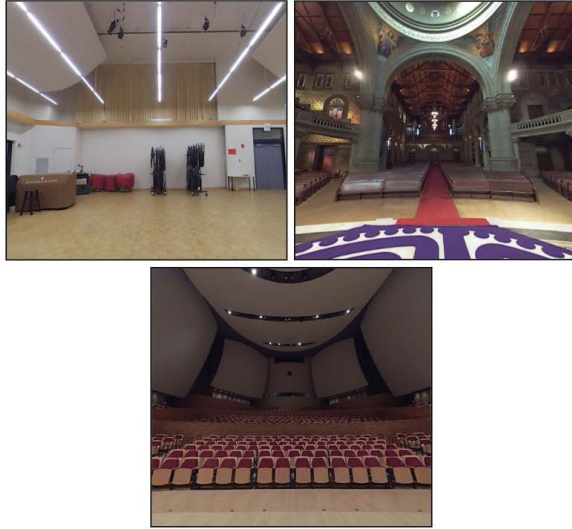


Figure 1: Screenshots of the 360° images within VR of the three performance spaces at Stanford University. top-left: Braun Rehearsal Hall, top-right: Memorial Church, bottom: Bing Concert Hall.

2 METHODS

2.1 Participants

Nineteen healthy volunteers (8 females and 11 males, $M(\text{Age}) = 27.63$, $SD(\text{Age}) = 8.64$) participated in the experiment. Participants were recruited through email soliciting keyboardists (requirement: minimum 5 years of keyboard playing). The experiment was conducted in a soundproof and acoustically-treated laboratory space at the Center for Computer Research in Music and Acoustics (CCRMA) at Stanford University. The study protocol was approved by Stanford University's Internal Review Board.

2.2 Stimuli and Apparatus

Performance Spaces

We used three music performance venues at Stanford University as environments for the experiment: Braun Rehearsal Hall, Bing Concert Hall, and Memorial Church (see Fig. 1). We chose these spaces because they have different acoustic characteristics and are commonly used for music practice and performance.

Room Acoustic Recording and Rendering

The impulse responses for the different spaces were recorded using a 10-second logarithmic sine sweep (sampling rate = 48k, frequency range: 20-24k Hz) played from a Bose SoundLink Revolve+ II Speaker (model no. 858366-1110). The sine sweep reflection was recorded 4 different times using a Zoom H3-VR (Item no: 450709) multi-channel recording device. The four takes were averaged together, deconvolved, and processed to remove noise to create a binaural impulse response audio file (see Table 1 for the reverberation times (RT60) and clarity indices (C80) of the different spaces). The dry impulse response was created in Matlab using the Signal Processing Toolbox [18]. We chose to present the audio binaurally because it is more accessible for a wider range of audiences compared to more involved approaches such as ambisonics and higher-order spatial audio rendering techniques, despite the latter providing higher resolution acoustic environment simulations.

Table 1: Reverberation time (RT60) and Clarity Index (C80) calculated from the impulse responses of the three experiment spaces.

Room	RT60 (seconds)	C80 (dB)
Braun Rehearsal Hall	2.65	5.45
Bing Concert Hall	3.60	2.29
Memorial Church	7.81	-2.33

Auditory Stimuli for VR

The auditory stimuli consisted of convolved keyboard piano sounds that the participants were triggering with the MIDI keyboard (M-Audio Axiom-61). We used the bitKlavier piano synthesizer [19] which reacts to velocity from the MIDI keyboard. The convolution of the piano sounds was done through Max/MSP (7.0.1, Cycling '74) using the HISSTools Impulse Response Toolbox [20] and presented binaurally through over-ear headphones (Sony MDR7506).

Visual Stimuli for VR

Visual conditions for the VR environments were created by taking 360° images using the Insta360 Pro Camera (UPC: 842126100208). The images were then edited in the Unity Editor to include a model of a keyboard, located in the exact location of the corresponding impulse response and facing the correct direction. The scores for the musical tasks were also added on a stand on top of the keyboard model. After constructing the scene, we used the 360° Screenshot Capture Asset from the Unity Asset Store to take a photo of the current camera view so that participants would all be facing the exact same way in the same position. The images were presented to the participants' HMD (Meta Quest 3) using the Pigasus VR Media Player.

The Musical Tasks in VR

We used two different scores for participants to perform in the different spaces and auditory feedback conditions (see Fig. 2 for the two scores). Score A contained only one level of rhythmic subdivision (quarter and eighth notes) and required slower finger movements to play the notes. Score B had two levels of subdivision (quarter, eighth, and sixteenth notes). This score was considered slightly more difficult and required faster finger movements to play the notes. Participants were given 4 counts (1 measure) of metronome ticks at 75 beats per minute (BPM) to prepare them to play the score, which stopped as soon as they started playing.



Figure 2: The two scores that were used during the experiment.

2.3 Procedure

Participants were seated in front of the MIDI keyboard in a soundproof room and were allowed to practice the two scores until they were comfortable with performing them accurately at 75 BPM. After that, the experimenter handed them the headphones (to put around their neck) and assisted them in putting on the HMD. After checking the orientation of the scene and the alignment of the real and virtual

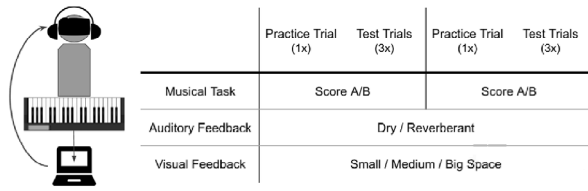


Figure 3: Left: The experimental setup. Participants were performing on a MIDI keyboard controlled by a Max/MSP patch running on a Mac laptop with the visuals presented through a head-mounted display. Right: An example of an experiment block showing practice and test trials starting with score A or B, then switching halfway through the block. Each block had dry or reverberant auditory feedback and one of the spaces presented visually. In the reverberant conditions, the auditory feedback matched the visual space. The order of the blocks was randomized for participants.

keyboards, participants were instructed to put on their headphones over the HMD. Participants were assigned a starting score and started with that score in each of the playing conditions. The assigned score was presented in the virtual scene on top of the virtual keyboard. Once the trials started, participants played 1 practice trial followed by three testing trials containing 3 repetitions of each score. Participants had to perform all repetitions without errors. Then they switched scores.

After the block was completed, participants took off their HMD and headphones to allow the experimenter to switch the visual scene to the next space. After performing in all the conditions, participants completed a post-test questionnaire regarding their experiences in VR and their sense of immersion in the environments. A custom-made Max/MSP program running on a Mac computer monitored the MIDI keyboard key-press timing, key-press velocity (sound intensity), and recorded the data into log-files. The same program also controlled the sound playback and convolution (see Fig. 3 for the experimental setup).

3 RESULTS

Immersion

After the experiment concluded, participants rated their level of immersion for the dry vs. reverberant conditions. A paired t-test revealed that the type of auditory feedback (dry vs. reverberant) had a significant effect on immersion ratings ($t(18) = -10.611$, $p < 0.0001$). Participants rated the reverberant conditions as more immersive than the dry conditions (see Fig. 4A). Participants were also asked whether they preferred the dry or reverberant conditions. The majority of participants preferred the reverberant conditions (80.95%) over the dry conditions (9.525%). The rest had no preference (9.525%) as shown in Fig. 4B.

Musical Behavior

We analyzed whether the type of auditory feedback affected participants' tempo and key-press intensity during the musical performance task. There were no overall generalizable differences in tempo between the dry and reverberant conditions common across participants. However, important differences emerged for the within-subject variability for dry vs. reverberant conditions for each participant. We computed the variability as the absolute difference between the mean values for the two conditions. A one-sample t-test revealed significant differences for both tempo ($t(18) = 3.54$, $p < 0.01$) and intensity ($t(18) = 5.8$, $p < 0.0001$) suggesting a change in homogeneity between conditions. Participants' playing speed varied more in the reverberant conditions compared to the dry conditions (Fig. 5A). However, for intensity measures, the opposite occurred as

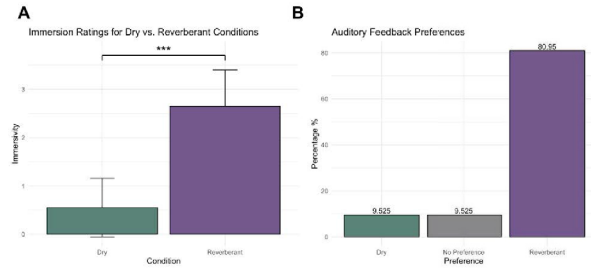


Figure 4: (A) Participants' immersion ratings for Dry (green) vs. Reverberant (purple) auditory feedback conditions. Ratings were on a scale from 0 to 5 with 0 being the least immersive and 5 being the most immersive. (B) The percentages of participants that preferred the dry or reverberant auditory feedback conditions, and those who had no preference for either condition (expressed in the gray bar in the middle).

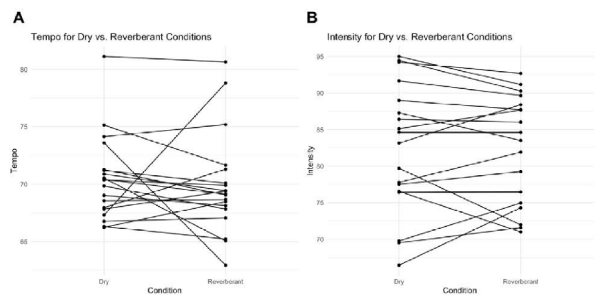


Figure 5: (A) The tempo variability between the dry and reverberant conditions for each participant. (B) The intensity variability of participants' key presses between the dry and reverberant conditions for each participant.

participants' overall key-press intensity became more homogeneous in the reverberant conditions (Fig. 5B).

4 DISCUSSION

In this study, we found that participants' musical behavior and sense of immersion were affected by whether or not they received auditory feedback with room acoustic cues while performing in VR-recreations of real spaces. Our results showed that the speed and intensity of participants' key presses were affected by whether they received dry or reverberant auditory feedback. Participants showed more variety in tempo in the reverberant conditions. This could be because they were trying to adjust to the performance space to enhance their musical expression through micro-timing. These results support previous findings that highlight solo musicians' individual approaches to performing in reverberant spaces [14].

In contrast, participants' key press intensity became more uniform in reverberant conditions. While these findings conflict with earlier research that reported no correlation between reverberation and note-press intensity [16], the increased similarity in intensity may reflect shared strategies to minimize the need for sound intensity adjustments. One possible explanation for this is that the prolonged note duration caused by increased reverberation time might enhance musicians' ability to hear and integrate auditory feedback, leading to more stable articulation. Alternatively, it could be due to the perceived loudness changes associated with the longer note durations introduced by the added reverberation.

Our questionnaires revealed that participants also preferred the reverberant playing conditions rated them as more immersive com-

pared to the dry conditions. Our data demonstrate the importance of adding realistic room acoustic cues to the auditory feedback of a virtual scene to ensure audiovisual cohesion. It is important to note that our virtual visual scenes are recreations of real spaces but only with basic features, leaving a lot of room for improvement to meet the industry-top standard of virtual recreations. Our motivation for using still images for the visuals is to put the main focus on the auditory feedback and how the addition of room acoustics can enhance a simple virtual scene.

In addition to using still images, we also did not animate or recreate participants' hands and piano key movements in VR. This was a limitation as some participants felt unnerved by not seeing their hands on their virtual keyboards in the environment and that detracts from the immersion and overall experience. We tried to minimize the need for participants to look at the keyboard or hands by keeping the musical scores simple and within a limited range of motion. We did, however, make sure that the virtual and real keyboards were aligned for the participant, and this was mostly successful according to the questionnaire feedback. Finally, our use of binaural feedback as opposed to ambisonics or higher-order spatial audio rendering techniques allows the experiment design to be easily reproducible for future music performance studies.

Our study was also limited to the VR apparatus and interface technology that we used. Participants noted experiences of mild discomfort in wearing the HMD and that it became heavy throughout the study, a challenge of using VR as a research tool in general. Interestingly, some participants commented that the metronome seemed to not belong to the space in the reverberant conditions as we used a dry metronome sound and did not convolve it for any of the conditions. This shows that participants preferred the reverberant sounds and were cognizant of the fact that the metronome sound was not part of the auditory scene experienced in the reverberant conditions.

5 CONCLUSION

In this experiment, we investigated how room acoustics affects musical behavior and sense of immersion in virtual-recreations of real spaces. We tasked participants to perform in both dry conditions (without room acoustics) and reverberant conditions (with room acoustics). Despite the simple rendering of the virtual spaces, our results indicate that room acoustics played an important role in how immersive participants perceived the scenes as well as how they performed a short musical piece. This underscores the importance of room acoustic cues in rendering virtual environments as it can affect the user's sense of immersion and their behavior. Further studies are needed to better investigate the effect of room acoustics on musical behavior and immersion in virtual spaces, especially using modeled spaces as opposed to still images.

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